

The Long Shadow of Servitude: The Causal Effect of 19th-Century Slavery on Contemporary Labor Market Inequalities in Brazil

Research Project

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“Economic Growth and Gender Dynamics in Historical Perspective”

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1 Introduction

Brazil was the largest importer of African slaves in the world, with approximately 5.5 million people coerced between the 16th and the 19th century (Eltis et al. 2021). It thus represents a very useful case study for understanding how slave-based extractive institutions in the past have affected economic conditions in the present. Using the Engerman and K. L. Sokoloff 1994 hypothesis as a theoretical framework, scholars have tried to understand the causal impact of Brazilian Slavery on inequality (Naritomi et al. 2012) or economic factors (Papadia 2019; Laudares et al. 2023). We contribute to this body of knowledge by exploring the effect of extractive institutions in Brazil on racial labor market gaps, as they have remained under-explored relative to broader inequality metrics and represent a crucial topic of current Brazilian policy-making. We further claim that the channel through which this effect takes place is through an unequal provision of human capital across the population. We test our theory by adopting an Instrumental Variable (IV) design. Using a measure of land suitability for the most slave-intensive plantations in Brazil and an index of proximity to the gold mines as instruments, we try to retrieve the causal impact of slavery presence on modern labor market inequalities. In order to ensure a spatially consistent comparison across time, we harmonize variables using the "AMC" methodology (Ehrl 2017). Our results seem to suggest that historical coercion shapes the quality and stability of insertion into the labor market rather than the decision to participate. At the same time, we found no evidence that the education inequality is the sole transmission channel affecting current outcomes, nor that is affected by slavery. Possible reasons might include that slavery affects inequality in terms of "quality" rather than quantity of education received. This suggests policies should target asset accumulation, spatial barriers, and educational quality, rather than solely increasing years of schooling. We also emphasize some limitations of the paper, more specifically concerning the weakness of the instruments and the harmonization process used.

The paper proceeds as follows: Section 2 provides an overview of the recent literature on the topic, Section 3 instead briefly summarises the history of slavery in Brazil. Section 4 refines theoretically the human capital mechanism as a transmission channel. Section 5 briefly introduces the data, while the methodology is defined in Section 6. Results are presented in 7, while Section 8 discusses the limitations of this paper. Section 9 concludes.

2 Literature review

A vast interdisciplinary literature examines the long-term consequences of slavery. Foundational frameworks by Engerman and K. Sokoloff 1994 and Acemoglu, Johnson, et al. 2001 argue that colonial labor systems, through factor endowments of the land, created extractive institutions and concentrated power, leading to durable developmental differences. Nunn 2008; Nunn 2007 extend this by documenting slavery's persistent effects on trust and institutional trajectories across Africa and the Americas.

Comparative studies reinforce these mechanisms. Buonanno et al. 2019 use historical slave shares to predict modern inequality in Colombia, while Acemoglu, García-Jimeno, et al. 2012 and Bertocchi et al. 2014 highlight similar human capital and discrimination channels.

In the Brazilian context, research highlights how slavery structured persistent racial hierarchies. Silva Jr. 2022 and Bucciferro 2017 show that these historical divisions remain evident in substantial racial gaps in education, occupation, and income long after abolition. Empirical work further

emphasizes local path dependence: Naritomi et al. 2012 link colonial structures to modern land inequality, while Papadia 2019 demonstrates that municipalities with higher historical slavery developed weaker fiscal capacity and worse public goods provision. Similarly, Laudares et al. 2023 confirms that areas more exposed to slavery exhibit weaker state capacity today.

Despite the breadth of this literature, the specific effects on contemporaneous labor market dynamics remain under-explored, notwithstanding recent studies documenting persistent disparities between the White and Afro-Brazilian populations in terms of wages, education, and occupational status (Reis et al. 2015). Understanding the historical roots of these gaps is crucial for the design of targeted policies, as disparities driven by deep-seated institutional inertia may require different interventions than those arising from contemporary market frictions. This is precisely what we are going to explore in this paper.

3 Historical Section

Throughout the period of Portuguese colonial rule (1500–1822), the administration established extractive institutions designed to harvest natural resources for the European market (Naritomi et al. 2012; Fausto 1999). As a result of this export-oriented model, Brazil became the largest importer of African slaves among the American colonies by 1640 (Laudares et al. 2023).

The initial phase of colonization, known as the “Century of Sugar” (1570-1670), saw sugar production and profits reach their zenith (Simonsen 1937). This commodity played a decisive role in shaping early settlement patterns, particularly in the Northeastern regions (Naritomi et al. 2012). In some states, slaves would constitute up to 65 to 70 percent of the population (Bethell 1987; Naritomi et al. 2012). While *Bahia* and *Pernambuco* served as the primary production centers, significant cultivation also occurred in *Rio de Janeiro*, *Espírito Santo*, *Paraíba*, and *Rio Grande do Norte* (Rout 1976).

Following the stagnation of the sugar economy, the discovery of alluvial gold in the late 1690s initiated the “Gold Cycle,” which dominated the 18th century and fundamentally shifted the colony’s economic and demographic axis from the Northeast to the South-Central interior, specifically in *Minas Gerais*, *Goiás*, and *Mato Grosso* (Boxer 1969; Fausto 1999). This extractive boom heavily intensified the demand for captive labor, with approximately 600,000 enslaved Africans being transported to the mining regions during the 18th Century (Bergad 1999).

By the 19th century, the economic dominance of the Southeast was further consolidated by the emergence of coffee as the primary export commodity. Major production centers emerged in *Rio de Janeiro*, *Minas Gerais*, and *São Paulo*, coinciding with the peak of the transatlantic slave trade as arrivals reached an all-time high between 1811 and 1856. Although the international slave trade was effectively abolished in the 1850s due to British pressure, the reliance of coffee plantations on slave labor remained undiminished (Costa 1982).

By the end of the 19th Century, the cessation of external slave imports necessitated a reallocation of the existing labor force across regions. While the total slave population began to decline, the inter-provincial slave trade allowed for a temporary increase in the slave population of *São Paulo* until such transfers were prohibited in the late 1870s. However, the inability to satisfy labor demand in the Southeast, compounded by high mortality rates, created conditions favorable to abolition. This political pressure, combined with slave revolts, ultimately led to the total abolition of slavery in 1888 (Papadia 2019).

4 Theoretical Framework

The persistence of slavery’s impact on modern racial labor market inequalities in Brazil can be traced to various channels discussed in literature. However, the one we find most convincing is the inter-generational transmission of human capital rooted in unequal public goods provision between white and Afro-brazilians.

This mechanism operates via the historical exclusion of the Afro-Brazilian population from public schooling. Slave societies created extreme inequality that allowed a small elite to capture political power (Engerman and K. L. Sokoloff 1994). In municipalities with a high historical presence of slavery, these elites had a strategic incentive to under-invest in public goods and to restrict access of mass education for the descendants of slaves in order to maintain a supply of low-skilled, immobile labor and to prevent their political mobilization (Bertocchi et al. 2014). This mechanism also took place in the Brazil, where literacy was a prerequisite for voting (Limongi 2016). Exclusion from public schooling and consequently being barred from accumulating human capital thus served as a mechanism of de-facto privation of rights. Moreover, because human capital is transmitted inter-generationally (Black et al. 2005), this initial exclusion created an educational deficit between white people and Afro-brazilians people (many of whom descendants of slaves) that persists today.

In the labor market this manifests as a wage gap, as the marginal productivity of black workers remains structurally lower than that of their white counterparts who benefited from better access to educational institutions (Goldin et al. 1999). Furthermore, this lack of skills restricts access to the formal sector, increasing the unemployment rate gap and reducing labor force participation among blacks facing high barriers to entry. This educational gap is also a primary driver of the informality gap. Formal sector jobs often require specific credentials or literacy levels that were historically denied to the black population. Without these qualifications, black workers are disproportionately pushed into the informal economy.

5 Data and Descriptive Statistics

This study will make use of granular Brazilian data, more specifically at the municipal level, in order to retrieve the causal effect of slavery in Brazil on current labor market inequality. We will link each municipality information from the 19th Century to the 2010 municipality data. To ensure a consistent spatial comparison across time, we will use the Minimum Comparable Areas (AMC) methodology by Ehrl 2017. Below will follow a brief description of the variables used, a description of the harmonization strategy and descriptive statistics (See Appendix B for a complete list of the variables used).

Dataset Description

Independent Variable: Slavery presence has been retrieved from the *1872 Brazilian Census* (IBGE 1874) as the ratio of slavery per population for each municipality at that time ¹.

Outcome variables: We used 4 proxies for labor market inequalities, all of them interpretable as a gap between people identifying themselves as “Branco” (white) and those identifying themselves as “Afro-Brazilians” (intended as aggregation of individuals identifying as “Preto” (Black) and “Pardo”

¹We would like to thank Papadia 2019 for providing the digitilised version of this census.

(Brown))². The proxies are the following: *Mincerian Earnings Gap* (hereby called Wage Gap), measuring the difference in log labor earnings between the above racial groups; *Informality Gap*, measuring the racial disparities in prevalence of employment without legal protection; *Labor Force Participation (LFP) Gap*, measuring the racial gap in labor market attachment; *Unemployment Rate Gap*, measuring the difference in unemployment rate between racial groups. To compute these labor market indicators, we utilized the harmonized micro data from the 2010 Census provided by Ruggles et al. 2025 in *IPUMS International 2025*. IPUMS aggregates smaller municipalities into spatially consistent units (GE02_BR) to ensure comparability between 1980 and 2010.³

To map this data to our unit of analysis, we first employed the official IPUMS correspondence table to link these harmonized units to the 2010 administrative municipality codes. While this procedure assigns the mean value of the IPUMS unit to its constituent municipalities, this concern is partially mitigated by our aggregation strategy. Since the historical Minimum Comparable Areas (AMCs) defined by Ehrl 2017 typically encompass outcomes from 1872 boundaries (which are significantly larger than modern municipalities), we argue that the IPUMS units should effectively be nested within the AMCs. Consequently, the intermediate “deharmonization strategy” that we used should not have an impact on the actual variation of the data at the level we are interested in. We test this argument in Appendix A. We observe that circa 76% of the IPUMS units is fully contained within a single AMC. This suggests that, while our argument holds for the majority of the case, there is still a share of IPUMS units whose municipalities are associated to different AMCs. Concerns of suppressed heterogeneity and induced measurement error are thus plausible.

Educational Inequality: To investigate the human capital accumulation channel, we construct a measure of racial educational disparity using microdata of *IPUMS International 2025* and elaborated by Ruggles et al. 2025 (considerations made in Appendix A apply). We calculate the *Educational Inequality Index* for each municipality as the difference in average years of schooling between White and Afro-Brazilian people. A higher value implies a higher degree of disparity in education across races.

Controls: A rich set of controls has been used in this study to account for pre-existing differences across municipalities that could independently drive both slavery presence and modern outcomes. These include a set of historical controls (retrieved from the Papadia 2019 digitized 1872 Census) and geographical characteristics constructed using QGIS relying on the shapefiles provided by Ehrl 2017. Further discussion on the sets of controls is made in the Section 6.

Instrumental Variables: As described in Section 6, we utilize two type of instruments, both connected to the historical determinants of slavery presence on Brazil (Section 3): suitability indexes for the most slave-intensive crops and an index of proximity to the gold mines. For the crops instruments, we utilize data on agricultural suitability from the Global Agro-Ecological Zones (*GAEZ v5*) database (*FAO*), and extract the land suitability indices for *sugar cane* and *coffee*. These indices measure the agro-climatic yield potential of the land based on soil and climatic characteristics, independent of current agricultural practices. The index runs from 0 (low suitability) to 1 (high suitability). For the second type of instrument, following the methodology of Naritomi et al. 2012, we create a measure of proximity to the gold mines in Brazil, I_m , defined as a function of the distance (in km) between a gold mine and a AMC centroid:

²This aggregation follows standard sociological and economic literature on racial inequality in Brazil, grouping *Pretos* and *Pardos* due to their shared socioeconomic disadvantages relative to the white population (Ministério da Igualdade Racial 2025).

³For details, see: https://international.ipums.org/international-action/variables/GE02_BR.

$$I_m = \begin{cases} \left(\frac{200-d_m}{200}\right)^2 & \text{if } d_m \leq 200 \text{ km,} \\ 0 & \text{otherwise.} \end{cases} \quad (1)$$

An index value of 1 indicates immediate proximity (effectively hosting a mine), while the value decays quadratically to 0 as the distance to the nearest mine reaches 200 km ⁴.

AMC Methodology and Harmonization procedure

As mentioned before, in order to ensure a spatially consistent comparison across time, we used the AMC harmonization procedure used by Ehrl [2017](#). This method resolves the issue of changing borders by creating spatially harmonized units. AMCs are constructed by aggregating modern municipalities to perfectly match the historical boundaries of the 1872 municipalities, allowing for a valid long-term comparison. More specifically, the study foresees the harmonization of 1872 variables, modern 2010 data and geographical characteristics.

1872 Variables We made use of the Crosswalk of municipalities between 1872 and 2010 provided publicly by Ehrl [2017](#). The data-file contains a list of AMC codes and associated to it a list of original names of municipalities in 1872 (when present) and the correspondent 2010 standard 7-digit IBGE code. This would then allow to merge the 1872 data using municipality name and then aggregating by AMC. We employed a two-stage matching algorithm to link the 1872 census municipalities to the AMC identifiers. In the first stage, we performed an exact match based on standardized municipality and province names. For the remaining unmatched observations, we implemented a fuzzy string matching procedure (using the Levenshtein distance algorithm) restricted within state boundaries to account for historical variations in spelling and to minimize false positives. We retained matches with a similarity score above 0.6 and manually validated the results. This procedure successfully matched the vast majority of historical municipalities to their corresponding AMCs. The final dataset contains 474 observations, slightly fewer than the theoretical maximum of 482 AMCs. This minor attrition occurs because certain AMCs correspond to regions that were either uninhabited or lacked administrative organization in 1872, such as parts of the Amazon and the western frontier, and thus do not appear in the historical census records. Importantly, checks confirm that no population centers from the 1872 Census were lost during the matching process, ensuring that the sample remains representative of the settled Brazilian territory at the time.

2010 Variables To harmonize the modern data with our historical unit of analysis, we utilized the aforementioned crosswalk. Since the 2010 datasets identify municipalities using the standard 7-digit IBGE code, we first merged the contemporary data with the crosswalk using this unique identifier. Subsequently, we aggregated all observations from the municipality level to the AMC level.

Geographical Controls Since physical geography is time-invariant, our harmonization strategy focuses on correctly assigning geographic features to the consistent 1872–2010 administrative boundaries. We computed all spatial variables directly at the AMC level using QGIS, utilizing the harmonized shapefiles provided by Ehrl [2017](#).

⁴Data on gold mine distributions were obtained from the (Serviço Geológico do Brasil [2025](#)).

Descriptive Statistics

Slavery Presence

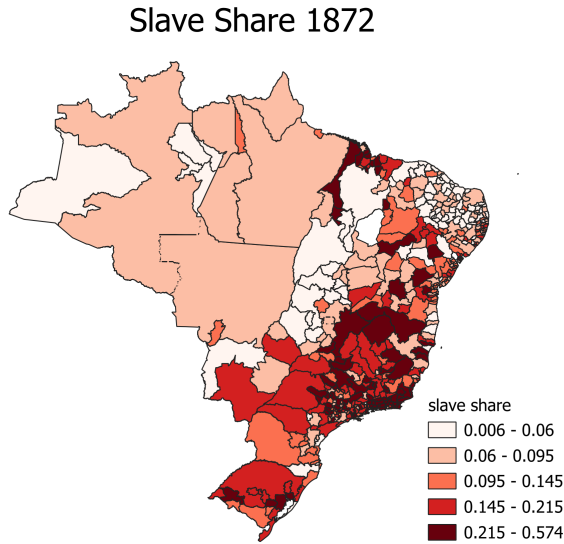


Figure 5.1: Historical slave share across Brazilian AMC units

Notes: This map visualizes the intensity of slavery across AMC units in Brazil. The “slave share” measures the proportion of the total population that was enslaved in 1872. Darker red areas indicate higher concentrations.

Sources: Slave share was retrieved from the 1872 Census given by Papadia 2019, whereas the shape-file is from Ehrl 2017. Elaborated by authors.

Figure 5.1 displays the spatial distribution of the enslaved population across Brazilian municipalities in 1872. We observe two distinct high-intensity clusters: the *South-eastern Paraíba Valley* and the *Minas Gerais* region. Some high intensity slave units can also be found in the Southern region of Brazil. Notice that, while some slave clusters are matched with suitable areas for extractive institutions (Figure 5.2), these distributions don’t fully match. This might be due to the temporal lag between the time slavery presence was recorded and the time the commodity cycles took place.

Crops Suitability and Gold Mines concentration

Figure 5.2 provides a spatial distribution of the agricultural suitability and gold proximity indexes. Suitability for coffee plantations seem to be more spread along the *Oeste Paulista* and *Zona da Mata* regions, where the crop flourished due to the presence of “purple earth” (Dean 1976). Conversely, sugarcane fertility traces along the Northeast coastline, along *Pernambuco* and *Bahia*. Finally, gold mines are most intensive in the area around *Minas Gerais* and the Northern part of Brazil.

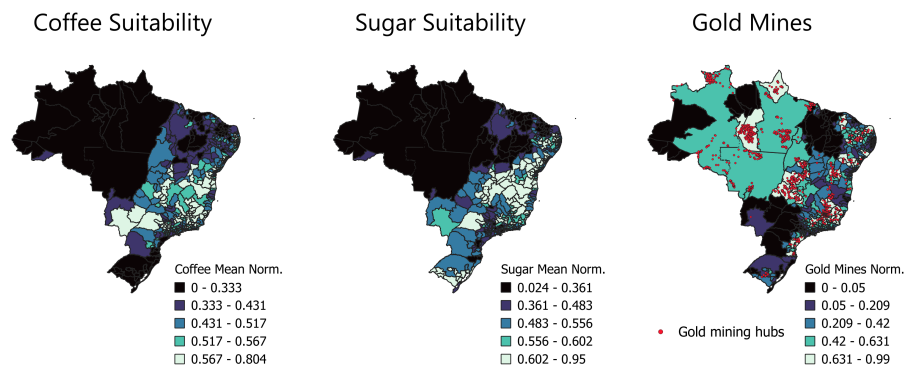


Figure 5.2: Coffee and Sugar suitability index distribution and gold mines concentration

Notes: These maps illustrate the geographical suitability for coffee and sugar canes and the proximity index to gold mines, for each AMC unit. All indexes run from 0 to 1, where the lighter the colors are, the more suitable the land/the higher concentration of gold mines. The red dots in the third map indicate the presence of a gold mine in the area.

Source: Raw Land Suitability data was retrieved from *GAEZ v5*, whereas data on the distribution of gold mines was retrieved from Serviço Geológico do Brasil 2025. The shapefile is still from Ehrl 2017. Elaborated by authors.

Labor Outcomes

Finally, Table 5.1 presents the summary statistics of our outcome variables. The data reveals persistent racial disparities in all dimensions of the labor market. The most pronounced inequality is observed in earnings, where White workers earn approximately 46% more than their Afro-Brazilian counterparts.

Quantity-based gaps are also evident, though smaller in magnitude than the wage gap. The Informality Gap averages 5.6 percentage points, indicating that Afro-Brazilians are systematically more likely to work without formal labor contracts. Similarly, the Unemployment Rate Gap shows that the unemployment rate for Afro-Brazilians is, on average, 2.5 percentage points higher than for Whites. These gaps are more pronounced along the South-Eastern coastline and the central region, around *Goiás* and *Minas Gerais* (Figure 5.3).

Table 5.1: Summary Statistics of Labor Market Outcomes

	N	Mean	St. Dev.	Min	Max
Wage Gap	482	.3835813	.148509	-.1149869	1.014692
Informality Gap	482	.0557847	.0322735	-.0700549	.1606945
LFP Gap	482	.013224	.0246741	-.0660317	.1137002
Unemployment Rate Gap	482	.0248755	.0197768	-.039399	.1228686

Notes: This table presents the summary statistics for the current (2010) labor outcomes for the harmonised AMC units.

Source: Data retrieved from Ruggles et al. 2025, elaborated by authors.

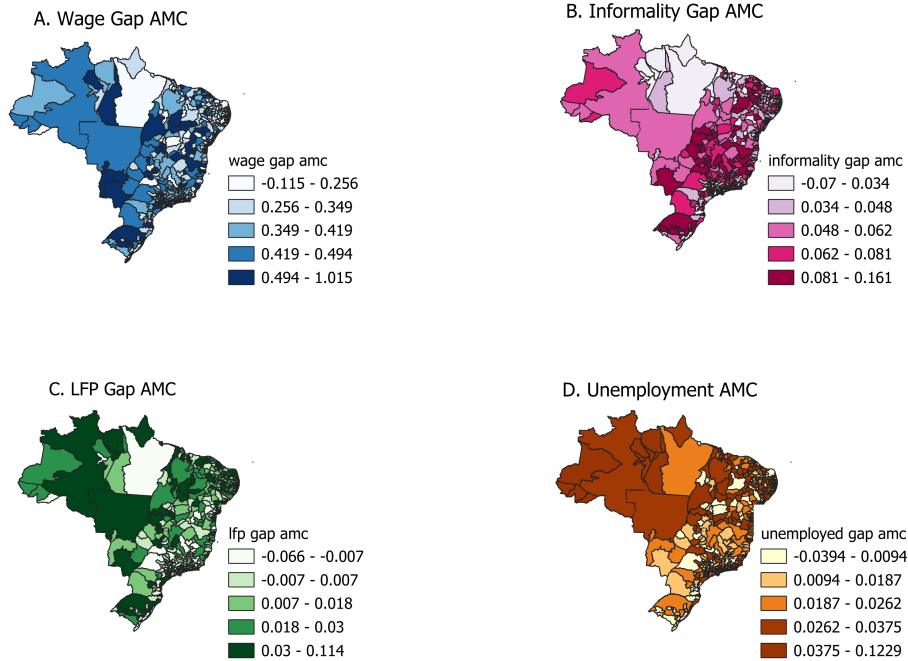


Figure 5.3: Modern Labor Market Outcomes (2010)

Note: These maps illustrate the geographical distribution of racial disparities in the Brazilian labor market. Map A refers to the Wage Gap, Map B to the Informality Gap, Map C to the Labor Force Participation Gap and Map D to the Unemployment Rate Gap.

Source: Data of labor outcomes is taken from Ruggles et al. 2025, whereas the shapefile from Ehrl 2017. Elaborated by authors.

6 Empirical Approach

This section discusses the econometric specifications of the study. We rely on cross-sectional variation across Brazilian AMCs to estimate the impact of slavery on modern labor racial gaps.

6.1 OLS Approach

To establish the correlational evidence, we first estimate a standard Ordinary Least Squares (OLS) model. For each AMC m , the relationship is defined as:

$$Y_m = \alpha + \beta_{OLS}S_m + \epsilon_m \quad (2)$$

where Y_m represents the labor market gaps. The variable of interest, S_m , is the slave share of the population in 1872.

We then add controls for time-invariant geographic characteristics as well as historical controls that may have jointly shaped colonial institutions and long-run development of each municipality. The equation then takes the following form:

$$Y_m = \alpha + \beta_{OLS}S_m + \mathbf{X}_m'\gamma + \epsilon_m \quad (3)$$

where $\mathbf{X}_m$ represents the vector of historical and geographical controls. Following Papadia 2019 and Laudaes et al. 2023, the historical controls include *population density*, *share of public employees*, *literacy rate*, *school attendance rate*, *share of free Afro-Brazilians*, *racial fractionalisation index*, *foreign share*, *gender ratio*, *agricultural* and *capitalist share*. In line with Naritomi et al. 2012, we initially included the following geographical controls: *latitude and longitude*, *distance from the Equator*, *distance from major cities*, *distance from the coast*, *altitude*, *ruggedness*, *climatic variability* and *general soil quality level* (see Appendix C for explanation). We acknowledge through diagnostic tests multicollinearity issues among some of these geographical measures, as detailed in Appendix D. However, we also show that the Variance Inflation Factor (VIF) for our main variable of interest (slavery) is only 2.31. This indicates that the multicollinearity is confined to the control set and does not inflate the standard errors of our variable of interest (Allison 2012). Including all controls should then effectively absorb geographical variation without compromising the precision of our main estimate. We nonetheless acknowledge that OLS estimates may still be biased by unobserved heterogeneity or measurement error. To address these concerns we introduce an Instrumental Variable (IV) strategy.

6.2 IV Approach

As mentioned above, using solely the OLS approach may still create endogeneity since there might be other unobserved factors correlated with slavery that affect current employment outcomes that we haven't controlled for. To name one, cities with better pre-existing road infrastructure in 1872 would have lower costs of transportation, facilitating the movement of slaves and potentially increasing slave density in the municipality, beside having a long lasting effect on the economic structure. If not properly controlled for, these effects could bias our estimates. Simultaneously, there might be measurement errors related to the 1872 Brazilian Census (Papadia 2019). As mentioned above, slavery in Brazil ended in 1888, thus it is extremely probable that the true distribution of slaves in Brazil during the 19th Century might have been quite different. For these reasons we employ a

two-stage least squares (2SLS) approach, where our instruments are the Land Suitability Indexes for sugar and coffee and a Proximity Index to gold mines.

The *First Stage* estimation retrieves the exogenous variation of slavery conditional on a set of controls $\mathbf{X}'_m\gamma$ and the 3 instruments ($\mathbf{Z}'_m$).

$$S_m = \alpha_1 + \mathbf{Z}'_m\beta_{1,IV} + \mathbf{X}'_m\gamma + \nu_m \quad (4)$$

We then proceed in the *Second Stage* to exploit the exogenous variation of slavery to retrieve its causal impact on current labor market outcomes.

$$Y_m = \alpha_2 + \beta_{2,IV}\hat{S}_m + \mathbf{X}'_m\gamma + \varepsilon_m \quad (5)$$

Given that all four dependent variables are defined such that higher values denote a greater disadvantage for the Afro-Brazilian population relative to the White population, we hypothesize the coefficient of interest to be positive ($\beta > 0$) across all specifications.

Control Choice This brief paragraph describes the different model specifications we are going to show in Section 7. We first are going to run a 2SLS regression using the same set of controls as in the OLS approach. However, we are aware that our crop based instruments are a non-linear function of a set of geographical controls such as temperature, soil quality and altitude. Including these variables as controls in our specification would then heavily reduce the variation left in the instruments to predict slavery, leading to a "Weak Instrument" problem. We therefore use a parsimonious set of controls and run a second 2SLS regression. This set is constructed to balance the need for variation in the instruments and to account for possible mechanisms through which the instruments could affect our dependent variable other than slavery presence (see Appendix E).

In order for these instruments to retrieve the true causal impact of slavery on current labor market outcomes, the following assumptions need to hold: the instruments must have a strong association with slavery presence (Relevance Assumption); the instruments must be uncorrelated with the structural error term of the outcome equation, conditional on covariates: $Cov(Z, \epsilon|X) = 0$ (Exclusion Restriction); monotonicity needs to hold.

Relevance Assumption We can prove this assumption by observing the *Kleibergen-Paap* F-Statistics to test for joint significance. Conceptually, this assumption should hold, since it is in line with Engerman and K. Sokoloff 1994's hypothesis, according to which factor endowments were fundamental for the creation of extractive institutions and the presence of slaves. Recent literature has also proved this hypothesis in other countries (e.g. Bertocchi et al. 2014, Acemoglu, García-Jimeno, et al. 2012) and Section 3 has already shown how strong the demand for slaves was for both plantations and gold mines. However, a potential concern might be the temporal lag between the peak of Brazil's extractive cycles and our measure of slavery intensity. The 1872 Census was conducted 16 years before abolition, long after the decline of the Sugar Boom and the Gold Rush. One might then worry that the correlation between historical suitability and the slave share in 1872 has been attenuated by centuries of economic adjustment, if not reversed. Furthermore, historical frictions in labor mobility, such as high transportation costs, may have prevented the spatial distribution of the enslaved population from aligning with the agricultural potential of the land or the proximity to gold mines. Technically, concerns may arise from the aggregation of the

land suitability data and from the mismatch between available gold mines in the 19th Century and those shown in the data. We will discuss this in Section 8.

Exclusion Restriction To interpret our estimates as causal, we assume that historical suitability for extractive institutions influences modern labor market gaps only through historical slavery. Firstly, we appeal to the “Structural transformation” argument, also highlighted by Bertocchi et al. 2014 in the US setting. Over the 20th century, the Brazilian economy transitioned from a totally agrarian export model to a service-and-industry-oriented economy. Given that agriculture accounts for a shrinking share of value-added in modern municipalities (*Censo IBGE 2010*), it is unlikely that historical soil suitability drives current labor market gaps through a direct effect on contemporary agricultural productivity. Similarly, the relevance of the gold mines in the Brazilian economy has long decayed (Papadia 2019), thus current municipalities’ outcomes should not be effected by how distant they are to mines. A second key argument exploits the distinction between inequality gaps and aggregate levels. Our dependent variables capture racial gaps in labor market outcomes, rather than overall income or development levels. While geographic endowments such as location or natural resources may directly affect aggregate economic performance, there is no clear mechanism through which our instruments should differentially penalize Black workers relative to White workers centuries later. As a result, any persistent effect of historical extractive suitability on racial labor market gaps is most plausibly transmitted through the racialized institutions created and reinforced by slavery. However, important threats to validity remain. For example, in the late nineteenth century, coffee-producing regions actively subsidized European immigration, often with the explicit goal of “whitening” the workforce (Sánchez-Alonso 2019). Those suitable areas that later received these high-human-capital migrants may therefore exhibit outcomes that reflect not only the legacy of slavery, but also subsequent demographic shocks that shaped labor-market structures.

Monotonicity This can be argued simply by stating that a more suitable land for plantation or gold mine presence shouldn’t induce the landowners to own less slaves than they would have otherwise had.

Education Channel Finally, we test whether educational inequality is the main channel through which slavery impacted current labor market dynamics.

We first create a separate Two-Stage Least Squares (2SLS) equation to determine if slavery causally predicts current educational stratification, in line with Bertocchi et al. 2014:

$$S_m = \alpha_1 + \mathbf{Z}_m' \pi + \mathbf{X}_m' \delta + \nu_m \quad (6)$$

$$I_m^{Educ} = \alpha_m + \beta_{Channel} \hat{S}_m + \mathbf{X}_m' \gamma + \varepsilon_m \quad (7)$$

The vector of controls $\mathbf{X}_m$ is the one of the parsimonious 2SLS specification. I_m^{Educ} represents the measure of educational inequality described in Section 5. A positive and statistically significant coefficient $\beta_{Channel}$ would confirm that historical slavery is a causal driver of contemporary racial disparities in human capital accumulation.

Then we proceed to test whether education is in fact a channel of transmission. We could estimate a simple OLS regression where we use the educational inequality index as a further control.

$$Y_m = \alpha + \beta S_m + \delta I_m^{Educ} + \mathbf{X}_m' \gamma + \epsilon_m \quad (8)$$

If education were to be an actual channel, we would expect the coefficient of slavery presence to drop with respect to our specification in Equation 3, as its association with labor market outcome should be controlled (at least partially) by the education variable. However, as evidenced by Becker et al. 2009, simply including a transmission channel as a control in a OLS specification does not fully resolve endogeneity issues. In this setting, both slavery and education inequality might in fact be endogenous. Therefore, in absence of an instrument for educational inequality, we implement a bounding analysis, where we obtain estimates of the effect of slavery on economic gaps net of the education effect. Instead of estimating the return to education endogenously, we impose a range of plausible returns derived from external literature to construct an "adjusted" dependent variable. We define the adjusted racial labor market gap (Y^{adj}) by subtracting the portion of the gap attributable to human capital differences from the observed gap (Y):

$$Y_m^{adj} = Y_m - (\rho \times I_m) = \alpha + \beta S_m + X_m' \gamma + \varepsilon_m \quad (9)$$

where I_m denotes our educational inequality index, and ρ represents the Mincerian return to schooling in log earnings. We do this in a 2SLS setting, where the *First Stage* is the same as Equation 4 and the *Second Stage* is:

$$Y_m^{adj} = Y_m - (\rho \times I_m) = \alpha + \beta \hat{S}_m + X_m' \gamma + \varepsilon_m \quad (10)$$

To calibrate ρ , we rely on established estimates for Brazil. We center our analysis on $\rho = 0.10$, consistent with the estimates from Psacharopoulos et al. 2018, while testing lower ($\rho = 0.08$) and upper bounds ($\rho = 0.15$) to account for historical variations. If the coefficient β remains positive and significant after this adjustment, the legacy of slavery may affect current labor market gaps through mechanisms beyond the educational attainment gap. It is important to note that this bounding analysis is restricted to the racial wage gap, as in the literature no comparable parameter exists regarding the marginal effect of an additional year of schooling on the probabilities of unemployment, labor force participation, or informality.

Note that all our regressions specification are conducted using robust standard errors in order to account for possible heteroskedasticity of the error term.

7 Results

7.1 OLS with and without controls

We begin the exposition of our findings by focusing on the OLS results. Table 7.1 shows that, in the uncontrolled specification, the coefficient on the Wage Gap is positive and highly significant (Figure F.1 shows this graphically). However, once controls are introduced, the coefficient turns negative and loses statistical significance. A similar pattern emerges for the Labor Force Participation and Unemployment gaps. This instability suggests that simple correlations between slavery intensity and modern labor outcomes are largely driven by omitted variables correlated with both historical slavery and present-day economic conditions.

Table 7.1: OLS with and without controls

	Wage Gap		Informality		LFP Gap		Unempl. Gap	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Share of Slaves (1872)	0.187*** (0.062)	-0.052 (0.090)	-0.002 (0.016)	-0.051** (0.024)	-0.068*** (0.010)	-0.016 (0.014)	-0.020** (0.008)	0.004 (0.012)
Constant	0.357*** (0.011)	1.243** (0.594)	0.056*** (0.003)	0.356*** (0.114)	0.023*** (0.002)	0.160* (0.090)	0.028*** (0.002)	0.077 (0.083)
Geographic Controls	No	Yes	No	Yes	No	Yes	No	Yes
Historical Controls	No	Yes	No	Yes	No	Yes	No	Yes
Channel Controls (1872)	No	Yes	No	Yes	No	Yes	No	Yes
R^2	0.017	0.196	0.000	0.171	0.080	0.258	0.011	0.177
Observations	474	474	474	474	474	474	474	474

Table 1 presents main results from the Baseline OLS regressions with and without the specified controls. The odd-numbered columns, (1), (3), (5), (7), present baseline estimates without controls serve as a benchmark, while the even-numbered columns, (2), (4), (6), (8), include a full set of geographic and historical controls. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

The explanatory power of the baseline OLS regressions is very limited. In the uncontrolled models, R^2 values are extremely low, ranging from 0.000 for the Informality Gap to 0.080 for the LFP Gap, indicating that historical slave share alone explains virtually none of the cross-municipality variation in contemporary labor outcomes. While the inclusion of controls mechanically increases R^2 , this reflects the addition of explanatory variables rather than genuine predictive power of slavery intensity itself. These results reinforce the concern that OLS estimates are biased and motivate the instrumental-variable strategy developed in the subsequent sections.

7.2 IV Results

We now turn to the instrumental-variable (IV) analysis. We first present a fully saturated specification that includes the complete set of historical and geographic controls as a diagnostic benchmark and then move toward more parsimonious specifications designed to restore first-stage relevance.

7.2.1 Full specification and 3 Instruments

The full-specification IV estimates that are reported in Table 7.2 includes the complete set of controls described in Appendix C. This specification is intentionally saturated and serves as a diagnostic benchmark rather than a preferred causal model.

The results highlight the severe consequences of over-saturation in IV estimation. The First Stage F-statistic is extremely low (0.57), far below conventional thresholds of 10 noted Stock et al. 2005, indicating a near-complete failure of identification. Notice moreover that since the agricultural instruments (coffee and sugar suitability) are constructed from climatic and geographic characteristics, simultaneously controlling for variables such as temperature seasonality, soil quality, and altitude absorbs almost all of the variation that would otherwise identify the endogenous regressor. As a result, these instruments retain virtually no independent predictive power (see Table F.1).

This lack of relevance leads to highly unstable and uninformative second-stage estimates. Although the point estimate for the Wage Gap is large in magnitude (2.164), its associated standard error (1.174) renders it statistically indistinguishable from zero. In this context, the coefficient should not be interpreted as evidence of a large economic effect, but rather as a manifestation of the well-known

Table 7.2: IV Estimates: Full Specification

	(1)	(2)	(3)	(4)
	Wage Gap	Informality Gap	LFP Gap	Unempl. Gap
Share of Slaves (1872)	2.164 (1.174)	0.025 (0.156)	0.038 (0.120)	0.290 (0.165)
Constant	3.948* (1.737)	0.449* (0.219)	0.226 (0.169)	0.426 (0.246)
Geographic Controls	Yes	Yes	Yes	Yes
Historical Controls	Yes	Yes	Yes	Yes
Observations	474	474	474	474
R^2	-0.820	0.145	0.236	-0.786
First Stage F-Stat	0.57	0.57	0.57	0.57
Hansen J P-Val	0.706	0.428	0.784	0.340

Notes: The table reports 2SLS estimates. The dependent variables are the racial gaps in wages (log difference), informality, labor force participation, and unemployment. All columns include the full set of geographic controls and historical controls from the 1872 Census. Robust standard errors in parentheses. Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

volatility of the 2SLS estimator under weak identification, where estimates exhibit heavy-tailed distributions and extreme sensitivity to sampling variation. Additional diagnostics reinforce this conclusion. The negative R^2 values indicate that the fitted values from the IV projection explain less variation than the sample mean, a common artifact when instruments are extremely weak. Similarly, although the Hansen J test does not reject the null of instrument validity, this result is not informative in the presence of such weak instruments. These findings demonstrate that the Full Specification suffers from severe multicollinearity and weak identification. Consequently, we do not view these estimates as economically meaningful.

7.2.2 Parsimonious Specification and 3 Instruments

We next report results from the parsimonious IV specification (Table 7.3). Relative to the Full Specification, this model restricts the control set to a selected group of core covariates (more in detail in Appendix E) in order to alleviate the severe multicollinearity and weak identification issues documented above. This parsimonious approach substantially improves identification strength. The First Stage F-statistic increases from 0.57 in the Full Specification to 4.86, indicating a partial recovery of instrument relevance. Nevertheless, this value remains below the conventional threshold of 10, placing this specification in the domain of weak instruments; moreover, sugar and coffee coefficients are statistically insignificant (Table F.3). Because of the weak instrument problem (as above) the Hansen J test is not informative.

Despite these limitations, the results provide economically meaningful patterns. The coefficient on the Wage Gap is positive and statistically significant (1.666), implying that a 10 percentage point increase in the historical slave share is associated with an approximately 16.7% increase in the modern racial wage gap. The reduction in the standard error relative to the Full Specification confirms that restricting the control set mitigates explosive standard deviations. We also find a positive and significant effect on the Unemployment Gap, suggesting that the legacy of slavery operates not only through wage penalties but also through persistent barriers to employment. In contrast, the coefficients for informality and labor force participation are small and statistically

Table 7.3: IV Estimates: Parsimonious Specification (3 Instruments)

	(1)	(2)	(3)	(4)
	Wage Gap	Informality Gap	LFP Gap	Unempl. Gap
Share of Slaves (1872)	1.666** (0.712)	0.027 (0.111)	-0.028 (0.086)	0.260** (0.106)
Constant	3.162*** (1.112)	0.487*** (0.163)	0.161 (0.134)	0.347** (0.176)
Core Controls	Yes	Yes	Yes	Yes
Observations	474	474	474	474
R^2	-0.460	0.140	0.244	-0.676
First Stage F-Stat	4.86	4.86	4.86	4.86
Hansen J P-Val	0.672	0.404	0.620	0.632

Notes: 2SLS estimates using Coffee, Sugar, and Gold suitability/proximity as instruments. "Core Controls" include longitude, distances to coast and major capitals, and 1872 historical covariates (density, literacy, sex ratio, foreign share, free black share). Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

insignificant, indicating that historical slavery does not primarily affect participation decisions or sectoral allocation, but rather outcomes conditional on labor market participation. Taken together, these findings motivate a further refinement of the identification strategy, which we implement by isolating a single, strong source of exogenous variation in the Gold-only specification.

7.2.3 Parsimonious Specification with Gold Instrument Only

In this final specification, we isolate exogenous variation of slavery to the gold index alone. This choice addresses the concern that our agricultural instruments are persistently weak. This, consistent with Bound et al. 1995, can produce finite-sample bias and unreliable inference.

The first-stage results reported in Table F.2 document a strong negative relationship between the Gold Index and slave intensity. This pattern might be explained consistently with Brazil's historical development, as the eighteenth-century mining boom had largely declined by 1872, while slave labor became increasingly concentrated in the expanding coffee frontier. This could mean that gold mines locations were purposely deprived of slaves in order to meet the demand for coffee plantations.

The results reported in Table 7.4 constitute our preferred 2SLS estimates. This choice is motivated by the strength of the first stage, as the Kleibergen–Paap F-statistic equals 12.49, exceeding the conventional rule-of-thumb threshold of 10 and alleviating concerns about weak identification (Stock et al. 2005). Accordingly, we interpret these coefficients as our most credible causal estimates. The wage-gap coefficient is positive and statistically significant (1.807). Because the dependent variable is a difference in log wages, this coefficient is a semi-elasticity: a 10 percentage point increase in the historical slave share is associated with an approximate 18.1% increase in the modern racial wage gap (F.O. approximation). For unemployment, the coefficient (0.268) is also positive and significant. Since the unemployment gap is measured in levels, a 10 percentage point increase in slave intensity widens the racial unemployment gap by approximately 2.7 percentage points. By contrast, the coefficients on informality and labor force participation are small and statistically indistinguishable from zero, suggesting that the slavery operates primarily through outcomes conditional on labor market participation rather than participation margins or informal allocation.

Table 7.4: 2SLS Results: Parsimonious Specification (Gold Only)

	(1) Wage Gap	(2) Informality	(3) LFP	(4) Unempl.
Share of Slaves (1872)	1.807** (0.793)	-0.013 (0.118)	-0.015 (0.091)	0.268** (0.113)
Constant	3.327*** (1.204)	0.441*** (0.171)	0.177 (0.138)	0.357* (0.182)
Core Controls	Yes	Yes	Yes	Yes
R^2	-0.573	0.158	0.245	-0.733
K-P Wald F-Stat	12.49	12.49	12.49	12.49
Observations	474	474	474	474

2SLS estimate using as instrument only the Gold Index and a parsimonious specification of controls. Robust standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

7.2.4 Robustness Analysis and Hypothesis Verification

A comparison across specifications is informative about the role of weak instruments. In the 3-instrument parsimonious model, the first-stage F-statistic (4.86) remained below conventional thresholds, raising weak-IV concerns. Weak-instrument theory predicts that 2SLS estimates are biased toward the OLS probability limit when instruments are weak (Stock et al. 2005). Since the OLS estimate with controls is close to zero ($\beta_{OLS} \approx -0.052$), attenuation toward zero is a plausible concern for the 3-instrument estimate (1.666). Consistent with this prediction, strengthening identification using the Gold-only instrument increases the wage-gap estimate from 1.666 to 1.807. While this comparison is not a formal test and may also reflect differences in the complier population across instrument sets, the monotonic increase is aligned with the direction of weak-IV attenuation. A similar, though smaller, pattern holds for the unemployment-gap coefficient.

7.3 Educational Channel

7.3.1 IV results with parsimonious specification

Table 7.5: Second Stage Education channel with 3IVs & 1IV

	Education Inequality Index (2010)	
	(1) 3 IVs	(2) Gold Only
Share of Slaves (1872)	1.674 (1.344)	1.505 (1.387)
Constant	5.016** (1.968)	4.818** (2.000)
Observations	474	474
R^2	0.027	0.056
First Stage F-Stat	4.86	12.49

Table 7.5 presents the "Parsimonious Specification" we have created for causal inference with 3IVs and only Gold. Column (1) includes coffee, sugar and gold as instruments, whereas the (2) column only gold. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7.5 shows the results of the 2SLS estimation of education gaps on slavery presence. The coefficients are positive, but not statistically significant, due to extremely inflated standard errors. This might suggest that historical slavery does not have a causal impact on educational inequality in terms of differences in educational years across ethnicities. Section 8 gives possible causes why this is the case.

7.3.2 Adjusted wage gap with 3IV and 1IV

Table 7.6 displays results of the bounded analysis for the adjusted wage gap with a parsimonious specification of controls, both with 3IVs and 1 IV. In both specifications, the coefficient of slavery presence remains positive and statistically significant across all tested values of ρ (0.08 to 0.15). This would imply that our hypothesis was not stable and there might be other channels affecting current labor market gaps other than educational inequality. At the same time, it is important to underline that the coefficients of this specification are slightly smaller than the results in Table 7.3. This might indicate that, while education is not the main transmission channel, controlling for it still reduces the impact of slavery on the outcomes.

Table 7.6: Adjusted Wage Gap with 3IVs and only Gold

	Adjusted Wage Gap							
	(1) 3 IV	(2) Gold Only	(3) 3 IV	(4) Gold Only	(5) 3 IV	(6) Gold Only	(7) 3 IV	(8) Gold Only
Slave Share (1872)	1.532** (0.645)	1.686** (0.726)	1.498** (0.630)	1.656** (0.712)	1.465** (0.615)	1.626** (0.698)	1.414** (0.596)	1.581** (0.678)
ρ	0.08	0.08	0.10	0.10	0.12	0.12	0.15	0.15
Observations	474	474	474	474	474	474	474	474
R^2	-0.47	-0.61	-0.47	-0.61	-0.47	-0.62	-0.45	-0.61
First Stage F-Stat	4.86	12.49	4.86	12.49	4.86	12.49	4.86	12.49

Table 7.6 presents the "Parsimonious Specification" we have created for causal inference with 3IVs and only Gold. The odd-numbered columns, (1), (3), (5), (7), present the estimates of the adjusted wage gap including all our instruments, (coffee, sugar, gold), while the even-numbered columns, (2), (4), (6), (8), only the gold proximity as instrument. We test for $\rho = 0.08$, $\rho = 0.10$, $\rho = 0.12$, $\rho = 0.15$.

Robust standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

8 Discussion

The results obtained in Section 7 can be summed up as follows: slavery persistently affects current wage and unemployment gaps; slavery has no significant impact on labor participation or informality; gold proximity is the only strong instrument; sugar and coffee indices prove weak; slavery doesn't explain education gaps; the bounding analysis suggests other transmission channels for slavery on current labor market gaps.

Notwithstanding some positive results, we find nonetheless important to stress some general concerns for our study. Firstly, slavery data is just a snapshot of 1872's demographics. This period was at the dawn of slavery abolition, thus it does not take into account the possible redistribution of slaves across the regions and across the centuries. This might have masked the true relationship between slavery presence and the location of extractive institutions. At the same time, 1872 was right before the mass immigration wave of skilled Europeans took place (Sánchez-Alonso 2019). This specialized type of immigrant was relevant for Brazil's industrialization, as skilled labor positions were directed to them rather than to Afro-Brazilians (Escamilla-Guerrero et al. 2024). This had a great effect on historical labor market dynamics, which could propagate until today. Endogeneity issues deriving from this are thus expected.

A significant challenge to the identification strategy might also internal migration flows, such as the movement of populations from rural to urban areas. If individuals moved systematically from regions with high historical slavery presence to more developed urban centers, these migration patterns could

confound the estimated impact of slavery on contemporary labor market inequalities. One potential, though data-intensive, solution to disentangle these effects would be to employ ancestry matching, which would allow for a more direct tracking of the causal link between historical servitude and modern outcomes across generations.

We also suspect suppressed variation in the data due to the aggregation methodologies we used. The AMC harmonization strategy detailed in Section 5, though useful to make temporal comparisons, heavily reduces variation in modern data, as it compresses observations units from over 5600 to around 500 (some of which of remarkable size in the Northern areas). This might have contributed to inflating the standard errors of our coefficients, leading to inconsistent results. This effect is to be compounded with the IPUMS “de-harmonization strategy” (see Appendix A). Given that the IPUMS units do not fully nest in one single AMC unit, some variation in current outcomes has been badly aggregated, thus leading to measurement errors and possibly biasing our results. The weakness of our land suitability instruments is another problem we faced. While we argued in Section 6 that these variables should have an effect on slavery presence because of the Engerman and K. L. Sokoloff 1994 hypothesis, we also mentioned possible concerns regarding the temporal lag between the time slavery data was captured and the commodity cycles phases. Aggregation problems are also a concern. The original indices are in fact calculated on a high-resolution raster grid, whereas our unit of analysis corresponds to relatively large Minimum Comparable Areas (AMCs). If plantation activity was concentrated in specific fertile enclaves within an unsuitable region, averaging suitability across a large AMC unit may have diluted the impact on slavery. Technical limitations may also be present for the gold mining instrument. Modern geological surveys identify deposits regardless of their historical accessibility. Including in the analysis deposits that were undiscovered at that time might then have created a measurement error.

The non-significance of slavery presence on LFP Gap and Informality Gap may be due to two concurring mechanism: a theoretical mechanism regarding participation and a methodological measurement error regarding informality. Historically, the transition from slavery to free labor did not always result in the withdrawal of the black population from the workforce, but rather their integration into the lowest rungs of the labor hierarchy (Bertocchi et al. 2014). Unlike wealthier segments of the population who may afford to delay entry into the labor market, descendants of enslaved people, historically denied asset accumulation or land property, are forced to participate in the labor market regardless of wage levels or working conditions. Consequently, the legacy of slavery would manifest not in whether Afro-Brazilians participate in the labor force, but in the quality of their insertion (e.g., higher unemployment risk and lower wages). The null result regarding the Informality Gap must also be interpreted with caution. Informality in Brazil is spatially concentrated in subnormal agglomerations (favelas), characterized by high density, irregular housing, and difficult access for Census personnel. Since moreover the Afro-Brazilian population is disproportionately represented in these informal settlements, the 2010 Census likely fails to capture the most vulnerable segment of the treated population. This systematic under-counting creates an attenuation bias, possibly underestimating informality gap and thereby masking the relationship between historical slavery and modern informal work.

It is also important to mention the insignificant effect of slavery presence on education years gap. The Brazilian Ministry of Education 2024 shows that years of schooling alone are insufficient to capture racial disparities in human capital formation. Non-white students are more likely to attend poorly resourced public schools and experience adverse school environments. Thus, even if years of schooling are not systematically different, descendants from slaves may have an inferior quality of education, which then impacts labor market outcomes. This could be tested by proxing educational

quality by *ENEM* results, a standardised university entrance exam which evaluates high school students in Brazil.

Finally, given the failure of our bounded analysis, we conclude that there might be complementary mechanisms through which slavery has a persistent effect on labor market gaps. A central channel operates through persistent wealth inequality. By preventing asset accumulation and inter-generational wealth transmission, slavery generated long-lasting differences in financial buffers and bargaining power across racial groups. Lower wealth constrains job search, weakens outside options, and increases vulnerability to unemployment shocks, even conditional on education. This mechanism provides a natural explanation for the large wage penalties and higher unemployment risk observed in historically slave-intensive areas, while remaining consistent with the absence of effects on labor force participation or informality (Darity et al. 2018). Spatial segregation constitutes a second key channel. Descendants of enslaved populations are more likely to reside in peripheral or poorly connected urban areas, where limited access to employment centers increases matching frictions and unemployment risk. Importantly, spatial mismatch can depress wages and raise unemployment without necessarily pushing workers into informal employment (Durlauf et al. 2013; Kain 1968). Finally, political under-representation may represent another institutional mechanism. Long-run exclusion from political representation may have persistently weakened Afro-Brazilian workers' access to job referrals and collective bargaining. This deficit in political and social capital reduces wage-setting power and employment stability even within formal labor markets, contributing to persistent racial wage and unemployment gaps (Putnam et al. 1993).

These channels suggest that history of slavery primarily affects the quality, stability, and remuneration of employment rather than extensive margins such as participation or formality. From a policy perspective, this implies that interventions focused solely on increasing education years are unlikely to close racial labor market gaps. Instead, policies should target asset accumulation, spatial barriers to employment, educational quality disparities and institutional mechanisms shaping bargaining power and access to high-quality jobs in historically disadvantaged communities.

9 Conclusion

Prior research has robustly established the long-run impact of slavery on institutional development and aggregate inequality. However, the specific transmission of these historical forces to contemporary labor market dynamics remained a critical knowledge gap. Our results address this by using 2SLS design with factor endowments proxies as instrumental variables. We find that slavery persistently widens wage and unemployment gaps but shows no significant causal impact on labor force participation or informality. This distinction suggests that historical coercion shapes the quality and stability of insertion into the labor market rather than the decision to participate. We also found that differences in years of schooling is not be the main channel of transmission, as difference in educational quality, political representation and spatial segregation might also play a role. We acknowledge some limitations, not on only methodological (see IV weakness) but also data-wise. We nonetheless believe in the usefulness of our research as a starting point for further studies. Using more reliable instruments to explain slavery or using alternative harmonization strategies (see Papadia 2019 and Quilombos), as well as relying on more advanced designs (e.g. spatial econometric designs to account for potential spillover effects) might effectively solve our issues and lead to a proper understanding of the slavery legacy on labor dynamics.

A AMC and IPUMS Harmonization: Comparison

Figure A1 illustrates the nesting structure of IPUMS units within historical AMCs from a graphical point of view.

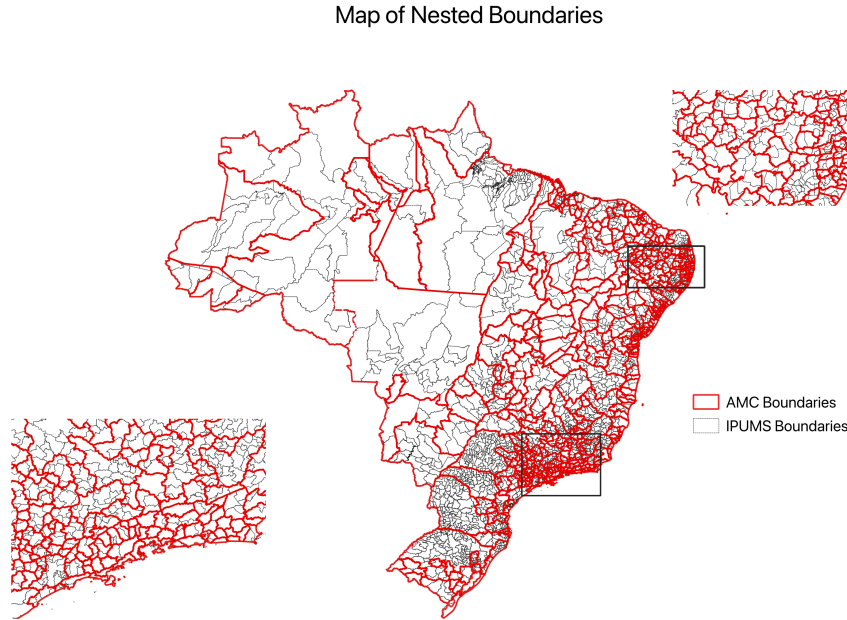


Figure A1: Nesting of IPUMS Harmonized Units within Historical AMCs.

Notes: The map displays the boundaries of the IPUMS harmonized geographic units (dashed lines) overlaid on the Minimum Comparable Areas (AMCs) defined by Ehrl [2017](#) (red lines).

Source: The shapefile of the AMC Boundaries comes from Ehrl [2017](#), whereas the IPUMS Boundaries shapefile was created by [IPUMS International 2025](#). The figure was created on QGIS by the authors.

Table A1: Distribution of Spatial Overlap (Nesting Consistency)

Nesting Category	Frequency	Percentage
Perfect nesting	4259	76.53
Minor overlap	1047	18.81
Moderate overlap	180	3.23
High overlap	68	1.22
Extreme overlap	11	0.20
Total	5565	100.00

Notes: Percentages refer to the share of IPUMS units in each nesting category. A value of 1 indicates perfect nesting (the IPUMS unit is fully contained within a single AMC).

Nesting Logic and Boundary Mismatches

Table A1 quantifies the spatial correspondence between the modern IPUMS harmonized units (2010) and the historical Minimum Comparable Areas (1872–2010).

- **Perfect Nesting (76.5%):** For three-quarters of the sample, the modern harmonization units fall entirely within the AMCs boundaries. In these cases, aggregating to the AMC level perfectly captures the local labor market variation without error.
- **Overlapping Units (23.5%):** The remaining units contain municipalities that belong to two or more different AMCs. In these instances, the value of the IPUMS unit (e.g., the mean wage) is assigned to all constituent municipalities, which are then aggregated into their respective historical AMCs.

The primary concern with overlapping units is the “smoothing” of variation across historical boundaries. If an IPUMS unit spans two AMCs, the distinct economic conditions of those two historical regions are averaged together in the modern data. Econometrically, this acts as a form of classical measurement error in the dependent variable. Standard econometric theory suggests that measurement error in the dependent variable inflates standard errors but does not bias OLS coefficients, unless the error is correlated with the independent variable. In our context, this implies that our estimates may be less precise, making it harder to find statistically significant results.

B List of Variables

Table B1: Description of the Variables

Main Variables	
<i>Wage Gap</i>	Difference in the natural logarithms of the mean wage between White and Afro-Brazilian group. Data from Ruggles et al. 2025, own calculation. Harmonisation performed as described in Section 5.
<i>Informality Gap</i>	Difference between Afro-Brazilian and White group in rate of informal workers (defined as "Employees without a legal contract") among employed workers. Data from Ruggles et al. 2025, own calculation. Harmonisation performed as described in Section 5.
<i>Labor Force Participation Gap</i>	Difference between White and Afro-Brazilian group in the Labor Force Participation rate. Data from Ruggles et al. 2025, own calculation. Harmonisation performed as described in Section 5.
<i>Unemployment Rate Gap</i>	Difference between Afro-Brazilian and White group in the Unemployment rate, defined as ratio of unemployed over the labor force participation. Data from Ruggles et al. 2025, own calculation. Harmonisation performed as described in Section 5.
<i>Slavery Presence</i>	Number of slaves in a municipality divided by total population. Data from IBGE 1874, thanks to Papadia 2019. Own calculation. Harmonisation performed as described in Section 5.
<i>Educational Inequality Index</i>	Difference in average years of schooling between White and Afro-Brazilian group. Data from Ruggles et al. 2025, own calculation. Harmonisation performed as described in Section 5.
Instruments	
<i>LSI Sugar</i>	Land Suitability Index for sugar, retrieved from GAEZ v5 , with settings low-input and rain-fed all seasons to mimic 19th Century's conditions faced by planters. Index running from 0 to 10000 normalised between 0 and 1. Own calculation using QGIS and performing zonal statistics on AMC shapefile from Ehrl 2017.
<i>LSI Coffee</i>	Land Suitability Index for coffee, retrieved from GAEZ v5 , with settings low-input and rain-fed all seasons to mimic 19th Century's conditions faced by planters. Index running from 0 to 10000 normalised between 0 and 1. Own calculation using QGIS and performing zonal statistics on AMC shapefile from Ehrl 2017.

Continued on next page

Table B1 – continued

<i>Gold proximity</i>	Index of proximity to gold mines. An index value of 1 indicates immediate proximity (effectively hosting a mine), while the value decays quadratically to 0 as the distance to the nearest mine reaches 200 km. Data on location of mines from Serviço Geológico do Brasil 2025 , distance calculated following methodology of Naritomi et al. 2012 using as a reference the centroid of each AMC unit. AMC shapefile from Ehrl 2017 . Own calculation.
Geographical Controls	
<i>Distance from Major cities</i>	Distance (in 100 kilometers) of the centroid of each AMC unit to the reference coordinates of the following cities: São Paulo, Rio de Janeiro, Salvador, Recife. Own calculations using QGIS based on data from IBGE 2023 and AMC shapefile from Ehrl 2017 .
<i>Distance from Equator</i>	Absolute distance between the AMC centroid's Y-coordinate and the Equator, in 100km. Own calculation using QGIS and AMC shapefile from Ehrl 2017 .
<i>Distance to the coast</i>	Linear distance from each AMC centroid to the nearest interpolated coastal point using the "Distance to Nearest Hub" algorithm, in 100 kilometers. Own calculation using QGIS and AMC shapefile from Ehrl 2017 .
<i>Latitude and Longitude</i>	The coordinates of each AMC's centroid, reconverted to decimal degrees for standard interpretation. Own calculation using QGIS and AMC shapefile from Ehrl 2017 .
<i>Altitude</i>	Average altitude in 100m of all elevation pixels within the AMC, using high-resolution raster data on altitude from AMBDATA 2025 . Own calculation using QGIS and AMC shapefile from Ehrl 2017 .
<i>Ruggedness</i>	Standard deviation of altitude within each AMC, using high-resolution raster data on altitude from AMBDATA 2025 , following work of Nunn and Puga 2012 . Own calculation using QGIS and AMC shapefile from Ehrl 2017 .
<i>Temperature Variability</i>	Standard deviation across the twelve monthly average temperatures for each AMC. Average gridded data from Berkeley Earth 2025 . Own calculation using QGIS and AMC shapefile from Ehrl 2017 .
<i>General Soil Quality</i>	We use raster data from the Harmonized World Soil Database (v1.2) (FAO 2025). The original data classify land into seven constraint categories, ranging from "No or slight constraints" (Class 1) to "Very severe constraints" (Class 4), as well as non-soil bodies (Classes 5–7). Following Nunn and Puga 2012 , we convert this categorical raster into a binary indicator equal to one for Class 1 and zero otherwise. We then compute zonal means within each AMC boundary.

Continued on next page

Table B1 – continued

Historical Controls	
<i>Population Density</i>	Natural logarithm of the ratio of the population in 1872 for each AMC unit by its area in square km. Population of 1872 harmonised at AMC level according to Section 5. Data from IBGE 1874 through Papadia 2019. Own calculation.
<i>Share of public employees</i>	Share of state related occupations in 1872 per total population. Data from IBGE 1874 through Papadia 2019. Harmonisation performed as described in Section 5. Own calculation. Harmonisation at AMC level according to Section 5.
<i>Literacy Rate</i>	Share of literate people among the free population. Data from IBGE 1874 through Papadia 2019. Own calculation. Harmonisation at AMC level according to Section 5.
<i>School Attendance Rate</i>	Share of people going to school considering the eligible share. Data from IBGE 1874 through Papadia 2019. Own calculation. Harmonisation at AMC level according to Section 5.
<i>Share of free Afro-Brazilians</i>	Share of free citizens of African descents (Black + Pardo) among the free population. Data from IBGE 1874 through Papadia 2019. Own calculation. Harmonisation at AMC level according to Section 5.
<i>Racial Fractionalisation Index</i>	Following Alesina et al. 1999, we created a measure of race distribution within each municipality. Data from IBGE 1874 through Papadia 2019. Own calculation. Harmonisation at AMC level according to Section 5.
<i>Foreign Share</i>	Share of non-African foreign citizens. Data from IBGE 1874 through Papadia 2019. Own calculation. Harmonisation at AMC level according to Section 5.
<i>Sex Ratio</i>	Sex Ratio among the total population. Data from IBGE 1874 through Papadia 2019. Own calculation. Harmonisation at AMC level according to Section 5.
<i>Agricultural Share</i>	Share of agriworkers among the total population. Data from IBGE 1874 through Papadia 2019. Own calculation. Harmonisation at AMC level according to Section 5.
<i>Capitalist Share</i>	Share of capitalists among the total population. Data from IBGE 1874 through Papadia 2019. Own calculation. Harmonisation at AMC level according to Section 5.

C Role of Specific Controls in our initial consideration

We include a comprehensive set of historical and geographic controls to isolate the causal effect of slavery from confounding factors such as initial institutional quality, economic prosperity, and geographic determinism. Below we detail the rationale for each included covariate.

C.1 Historical Controls

To rule out the *persistence of initial wealth* and demographic composition as mechanisms, we control in our model specifications for municipality characteristics observed in 1872 (harmonised at the AMC level). This ensures that our results capture the institutional legacy of slavery rather than the persistence of pre-existing economic, human capital, or racial structure advantages.

- **Human Capital (Literacy and School Attendance):** We control for the *literacy rate* and *school attendance rate* of the free population in 1872. This accounts for initial differences in human capital accumulation, ensuring that modern labor market outcomes are not simply a function of early educational advantages in certain regions.
- **Population Density:** Used as a standard proxy for initial economic development and urbanization in 1872 (Acemoglu, Johnson, et al. 2001). This control allows us to compare municipalities that had similar levels of economic activity and settlement density in the 19th century but differed in their reliance on slave labor.
- **Share of Free Afro-Brazilians:** Controls for the presence of a free black and mulatto population in 1872. This is crucial to ensure that our results capture the impact of coercion (slavery) rather than the general presence of an Afro-Brazilian population, which may face discrimination regardless of legal status.
- **Racial Fractionalization Index:** Constructed following the methodology of Alesina et al. 1999 this index measures the probability that two randomly selected individuals from the municipality belong to different racial groups. This controls for the initial ethnic distribution of the population within each municipality, which might have an impact on the initial provision of public goods.
- **Share of Foreigners:** Controls for the openness to immigration and the presence of European settlers. Since European immigrants brought distinct skills and institutions, omitting this variable could confound the effect of slavery with the effect of a smaller immigrant population.
- **Sex Ratio:** Defined as the ratio of men to women. This controls for the demographic structure of the municipality, distinguishing between settled agricultural societies (balanced ratios) and unstable frontier or mining towns (high male ratios).
- **Share of Public Employees:** A proxy for *state capacity* and the presence of colonial administration. This controls for the reach of the central government and legal institutions, which correlates with both slavery and modern development.
- **Economic Structure (Agricultural and Capitalist Shares):** We include the share of the workforce employed in agriculture and the share of "capitalists" (property owners/merchants). These variables control for the initial structural composition of the economy, distinguishing between purely agrarian societies and those with an emerging commercial class.

C.2 Geographic Controls

To rule out *geographic determinism*, we include controls that account for climatic, topographical, and soil features correlated with economic outcomes.

- **Climate (Temperature Variability, Latitude and Distance from Equator):** These variables capture the broad climatic and geographic gradients that characterize Brazil’s economic geography. We control for *temperature variability* to distinguish between the constant warmth of tropical regions and the seasonal cycles of subtropical areas. Literature suggests that seasonal temperature variation is positively correlated with economic outcomes, as the absence of frost in the tropics facilitates higher disease burdens and pest persistence, limiting productivity (Masters et al. 2001; Sachs 2001).

Furthermore, we include *latitude and distance from the equator* to account for the well-documented “tropical disadvantage.” Empirical evidence consistently shows a robust correlation between distance from the tropics and development, driven by channels such as agricultural technology transfer constraints, disease environments, and the historical formation of institutions (Gallup et al. 1999; Dell et al. 2012).

- **General Soil Quality:** Following Nunn and Puga 2012, we control for the natural fertility of the land using the *share of land with high nutrient availability* (derived from the Harmonized World Soil Database). This control is essential to distinguish the effect of generally fertile land, which could drive prosperity and affect current economic outcomes.
- **Ruggedness (Altitude Mean and Standard Deviation):** Terrain ruggedness historically influenced transport costs, the feasibility of plantation agriculture, and the enforcement of slavery. By controlling for altitude and its variance, we isolate the institutional channel from physical geography barriers.
- **Market Access (Distance to Coast and Longitude):** Coastal municipalities historically had better access to international markets and trade. We control for distance to the coast to ensure that our estimates do not merely reflect the advantage of proximity to ports.
- **Distance to Economic Hubs:** We include the geodesic distance to the four major historical and economic capitals: São Paulo, Rio de Janeiro, Salvador and Recife. This controls for gravity effects and economic spillovers from major agglomerations, which strongly influence modern labor market dynamics.

D Multicollinearity Issues

As mentioned in the main paper, we test for multicollinearity among the geographical variables.

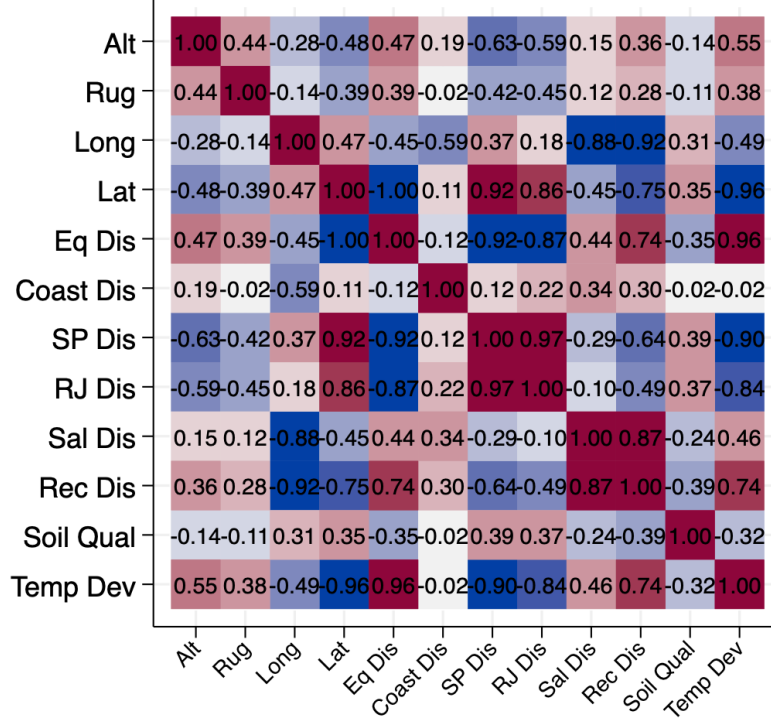


Figure D.1: Correlation Matrix of geographical variables

Notes: This figure shows the correlation matrix of all geographical variables. More intense colors indicate a higher correlation index, with red for positive sign and blue for negative. **Sources:** Elaborated by authors.

As visible in Figure D.1, we have the following multicollinearity clusters:

- Latitude, Distance from the Equator, Temperature Variabilty and Distance from São Paulo.
- Longitude, Distance from Recife and from Salvador.

The first cluster arises from the predominantly vertical orientation of the Brazilian territory. Since the country lies almost entirely in the Southern Hemisphere, an increase in absolute latitude is equivalent to an increase in the distance from the equator. This southward progression correlates with climatic patterns, specifically the higher temperature seasonality found in subtropical regions compared to the tropical north. Furthermore, because São Paulo is located in the southern portion of the country, moving south simultaneously reduces the distance to this economic hub while increasing latitude and temperature variability.

The second cluster reflects the horizontal expansion of Brazil from the Atlantic coast toward the interior. Salvador and Recife are situated on the northeastern coast. Consequently, longitude is basically a linear proxy for the distance to these historical centers. As one moves inland toward the west, longitude changes strictly alongside the increasing distance from both Salvador and Recife, creating a mechanical correlation between the geometric coordinate and these specific distance measures.

Table D.1 quantifies this pattern using the Variance Inflation Factor (VIF). Formally, the VIF for a given predictor j is defined as:

$$VIF_j = \frac{1}{1 - R_j^2}$$

where R_j^2 is the coefficient of determination obtained from regressing predictor j on all other covariates. A high VIF indicates that a variable shares a substantial amount of variance with other predictors, which mechanically inflates the standard errors of that specific coefficient (Wooldridge 2019; Greene 2018). Consistent with the clusters identified above, the geographical covariates exhibit extremely high VIFs, implying high uncertainty in the estimation of their individual partial effects. However, the VIF for our variable of interest (slave share) remains very low (2.31).

We therefore decide to retain the full set of geographical controls. Since the multicollinearity seems to be confined to the control variables, it should not contaminate the precision of the slavery coefficient (Allison 2012). On the contrary, removing these controls would essentially trade a loss of efficiency for a loss of consistency. Since geographical factors such as distance to the coast or climatic suitability likely determined both historical slavery placement and contemporary economic outcomes, omitting them would trigger omitted variable bias. Formally, if a relevant geographic control Z is excluded, the estimated coefficient for slavery ($\hat{\beta}_{slavery}$) is:

$$E[\hat{\beta}_{slavery}] = \beta_{slavery}^{true} + \beta_Z^y \cdot \frac{Cov(Slavery, Z)}{Var(Slavery)}$$

Given the non-random distribution of slavery across Brazil’s geography, the covariance term is likely to be non-zero. Thus, dropping these controls would absorb their effects into the slavery coefficient, biasing our main estimate. We therefore prioritize structural validity over the reduction of multicollinearity among controls (Wooldridge 2019).

Table D.1: Variance Inflation Factors

	VIF
Latitude	3166.79
Longitude	180.15
Equator Distance	3019.09
Recife Distance	274.93
RJ Distance	144.21
SP Distance	132.47
Salvador Distance	19.36
Temp Variability	21.03
Coast Distance	9.25
Altitude	3.18
Ruggedness	1.70
Black Free Share	2.80
Slave Share	2.31
Pop. Density	2.23
Foreign Share	1.74
School Attendance	1.52
Racial Fract.	1.51
Soil Quality	1.42
Sex Ratio	1.31
Literacy Rate	1.31
Agri Share	1.29
Public Share	1.18
Capitalist Share	1.12
Mean VIF	304.00

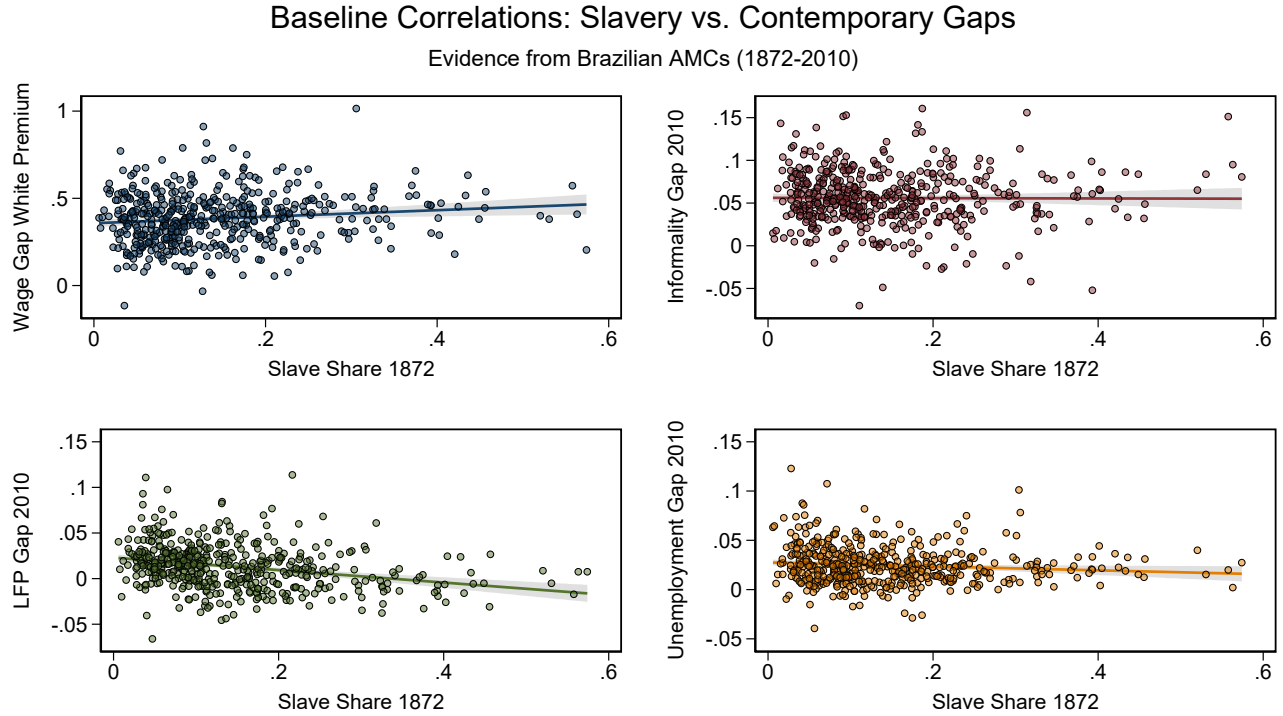
Notes: This table shows the VIF values of our analysis, both with the full set of geo controls and the parsimonious one. It is important to note that the Variance Inflation Factor is an intrinsic property of the matrix of regressors and is mathematically independent of the outcome variable. Consequently, the VIF diagnostics reported here apply identically to all regression specifications in the analysis (Wage Gap, Informality, LFP, and Unemployment) that utilize the same set of control variables. **Sources:** Elaborated by authors.

E Parsimonious IV specification

In order to avoid the “Weak Instrument” Problem in our 2SLS regression, we use a parsimonious set of controls, constructed in such a way to balance the need for variation in the instruments and to account for possible mechanisms through which the instruments could affect our dependent variable other than slavery presence. In order to do so, we dropped controls which we think heavily correlate with our instruments. These are:

- **Temperature Variability and Latitude:** We exclude these variables because they are the fundamental climatic determinants used to construct the Land Suitability Indexes for coffee and sugar. According to the FAO GAEZ methodology, crop suitability is non-linearly determined by thermal regimes and photoperiods, which are direct functions of latitude and temperature variability (Fischer et al. 2002). Including these as controls would result in near-perfect multicollinearity with our instruments, absorbing the specific variation in crop potential we aim to exploit and severely reducing the power of our first stage.
- **Altitude and Ruggedness:** We omit altitude and ruggedness to preserve the relevance of the instruments. Coffee suitability is mechanically determined by elevation, thriving specifically in high-altitude zones (1,000–2,000 meters). Similarly, geological gold deposits are frequently located in orogenic belts associated with rugged terrain. As noted by Nunn and Puga 2012, terrain characteristics often dictate the location of resource extraction; thus, controlling for them would effectively control for the instrument itself.
- **General Soil Quality:** We exclude general soil quality because it is a primary input in the calculation of agro-climatic yields. The suitability instruments are composite indices that already account for soil constraints (nutrient retention, rooting conditions, etc.). Controlling for the raw soil quality would result in “over-controlling,” where the control variable captures the same information as the instrument, leading to a loss of identification variation (Costinot et al. 2016).

F Tables and graphs



Note: Plots show linear fit with 95% confidence intervals.

Figure F.1: Baseline Correlations: Slavery 1872 vs. Contemporary Labor Gaps 2010

Note: This figure presents unconditional correlations between historical slavery intensity in Brazil based on the census 1872 (through Papadia 2019) and contemporary labor outcome gaps at the AMC level in 2010. Each dot represents a AMC unit.

Source: Data of labor outcomes is taken from Ruggles et al. 2025, whereas Slave Share from IBGE 1874.

OLS Scatterplots Figure F.1 displays unconditional correlations between slavery intensity in 1872 and contemporary labor market gaps. The wage gap scatterplot shows a mild positive association, consistent with the positive OLS coefficient in the uncontrolled regression. In contrast, the relationship between slavery and informality appears weak and largely flat. The bottom panels indicate modest positive correlations between historical slavery and both labor force participation and unemployment gaps. However, the substantial dispersion of observations and the sensitivity of these correlations to the inclusion of controls suggest that these patterns should be interpreted as descriptive rather than causal.

Table F.1: First stage Regression with 3 IVs (Full Specification)

	(1) Slave Share 1872
Coffee Suitability	-0.027 (0.035)
Sugar Suitability	0.025 (0.034)
Gold Index	-0.030** (0.012)
Constant	-1.102*** (0.275)
R-Squared	0.573
F-Statistic	23.04
Observations	474

This Table presents the first stage of the 2SLS regression with the "Full Specification". We use all the instruments: coffee, sugar and gold indexes. Robust standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table F.2: First Stage Regression with 1 IV (Parsimonious specification)

	(1) Slave Share 1872
Gold Index	-0.041*** (0.011)
Constant	-1.027*** (0.255)
R-Squared	0.529
F-Statistic	25.79
Observations	474

This table presents the first stage of the 2SLS regression with the "Parsimonious Specification" using only 1 Instrument. Robust standard errors in parenthesis.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table F.3: First stage Regression with 3 IVs (Parsimonious Specification)

	(1) Slave Share 1872
Coffee Suitability	0.001 (0.034)
Sugar Suitability	0.027 (0.036)
Gold Index	-0.042*** (0.012)
Constant	-1.012*** (0.257)
Observations	474
R-Squared	0.530

This Table presents the first stage of the 2SLS regression with the "Parsimonious Specification". We use all the instruments: coffee, sugar and gold indexes. Robust standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table F.3 shows that the coefficients for coffee and sugar are very low and statistically not significant predictors of the Share of Slaves, whereas the coefficient of gold proximity is negatively associated with slave share and is significant at the 1% level of significance. We can argue the negative relationship based on the fact that Brasil was already exposed to gold mining almost two decades before the arrival of slaves there. When the enslaved population arrived in Brazil the massive production of sugar and coffee had already taken the place of gold minings.

G AI Usage - Gemini 3.1 Pro Prompts

In the spirit of transparency, we adhere to a policy of full disclosure regarding the use of Artificial Intelligence (Model Gemini 3.1 Pro) in the preparation of this proposal. We utilized AI tools strictly as auxiliary mechanisms to enhance technical efficiency and linguistic clarity, without replacing critical reasoning. Specific prompts used are listed below.

Code Debugging for Data Preparation and Analysis

Please help identify the problem in the following lines of code:

[Area where specific code lines and the corresponding error message was provided.]

This prompt was used to debug code segments that did not initially work as intended during the harmonization procedure and the generation of plots and regression results.

LaTeX Formatting

Please typeset the following equation/line in LaTeX:

[Area where the text or equation was provided.]

There is an error in the LaTeX equation. Please help to solve the error.

Please transform our regression results *[screenshot of the regression table]* into a shorter and cleaner table.

How to link code-generated plots into LaTeX?

The tables take up substantial space in our paper, and we are constrained by a page limit. How to arrange these plots so as to adjust their size? *[Area where existing code lines for plotting were provided.]*

Please typeset the following citation in LaTeX format for the bibliography section of the paper: *[Area where the citation was provided.]*

These and similar formatting-related prompts were used whenever we needed assistance converting specific elements into LaTeX format.

QGIS Navigation

Please explain how to perform a *[Specific QGIS function]* on this shape-file of Brazil.

Language Polishing

Please review the following text for grammar, clarity, and flow. Revise for readability while preserving the original meaning and academic tone. *[Area where the text was placed. Short excerpts (usually no more than 2–3 sentences at a time) from selected parts of the paper were provided.]*

This prompt was used consistently during the editing process to improve the clarity and readability of selected sections of the text. The authors confirm that the intellectual core of this proposal, including the research questions, methodology, and data analysis, is entirely human-generated. We guarantee that no part of the dataset was synthetically created or augmented by AI.

H Working division

The following section describes how the making of this research project was made, in terms of work division and responsibility.

- **Idea:** Federico Proverbio
- **Literature Review:** Federico Proverbio 25%, Eirini Lontou 25% Federico Stoppa 25% Nishant Jha 25%
- **Theoretical Framework:** Federico Eolo Otto Stoppa
- **Data Cleaning and Preparation:** Federico Proverbio 40% Eirini Lontou 25% Federico Stoppa 30% Nishant Jha 5%
- **Harmonization Procedure:** Federico Proverbio
- **Historical Section:** Nishant Jha
- **Descriptive Statistics:** Eirini Lontou
- **Theoretical Analysis:** Federico Proverbio 50%, Federico Stoppa 50%
- **Results:** Federico Stoppa 60%, Eirini Lontou 40%
- **Discussion:** Federico Proverbio 50%, Eirini Lontou 50%
- **Appendices:** Federico Proverbio

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